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Laboratory cooking and degradation experiments to support interpretations of absorbed lipid residues arising from the processing of broomcorn millet (*Panicum miliaceum*) and animal products in archaeological pottery vessels I. Molecular analysis ☆

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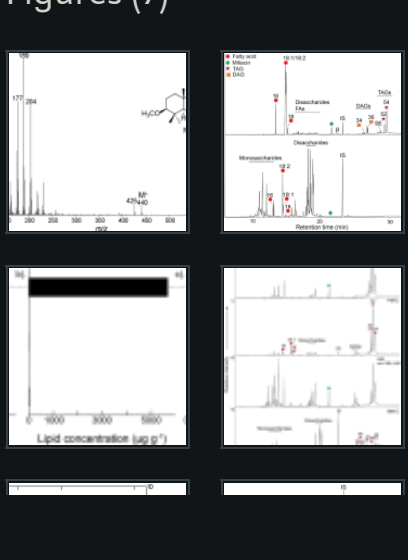

Highlights

- Stimulation of broomcorn millet and millet-meat cooking and decay in replica pots.
- Co-released animal fats enhance miliacin incorporation into replica pots.
- Miliacin survives better than fatty acyl lipids in decay experiments.
- Co-occurrence of miliacin and fatty acyl lipids reveals millet-animal processing.

Abstract

Plant resources were essential in prehistoric diets and their processing in pottery vessels, especially of grains, represented a crucial innovation. Lipid residue analysis of unglazed ceramics is a valuable tool for identifying ancient dietary components. While animal lipids are by far the most widely detected, increasing studies show their co-occurrence with plant lipids derived from grains or leaves. A notable observation is the broomcorn millet triterpenoid biomarker, miliacin (olean-18-en-3β-ol), occurring together with acyl lipids dominated by palmitic ($C_{16:0}$) and stearic ($C_{18:0}$) acids in archaeological pottery. The co-occurrence of these lipids of contrasting sources and structures complicates the interpretation of the contributing components of mixed residues due to the potentially (i) differential release and deposition rates of cereals *versus* animal lipids, and (ii) contrasting survival rates of triterpenoids *versus* acyl lipids during burial. To resolve these questions laboratory cooking and degradation experiments were conducted involving millet alone and millet/animal product mixtures. Only low concentrations of millet lipids were incorporated in replica pottery when the grains were boiled alone. Co-processing with meat enhanced miliacin incorporation, although the millet acyl lipids were completely masked by animal fats. Degradation experiments showed miliacin is highly resistant to microbial degradation. After extended degradation, the residues from the sherds dosed with millet lipids were predominately composed of miliacin, with near complete loss of acyl lipids. The degraded residues of mixed millet and animal lipids were dominated by $C_{16:0}$ and $C_{18:0}$ fatty acids alongside miliacin, suggesting that co-processing of animal products and millet would be detectable in archaeological pottery.

Keywords

Lipid residue analysis; *Panicum miliaceum*; Miliacin; Pottery; Cooking; Degradation

Previous article in this issue

1. Introduction

Plant resources such as grains, leaves, fruits and nuts were widely exploited in prehistory as dietary staples (Zeder, 2012). Although some plant products can be consumed raw, processing, especially of grains and seeds, was a crucial innovation to maximise energy gains, improve palatability, and remove toxic substances (Carmody and Wrangham, 2009). However, discerning food processing practices from the archaeological record is challenging. While archaeobotanical and zooarchaeological remains can provide insights into the range of commodities exploited, many plant species and animal products, such as milk, are invisible in the archaeological record. The analysis of organic residues preserved in unglazed archaeological pottery vessels has been developed as a valuable alternative approach for recovering indicators of dietary foodstuffs. Lipid residues trapped within the porous ceramic matrix are particularly well-preserved due to their inherent hydrophobicity arising from their predominantly saturated hydrocarbon structures. The trapping within pores or adsorption within the clay microstructure reduces lipid losses by limiting microbial degradation during burial.

The identification of absorbed lipids, and their assignment to specific commodities, relies on detailed knowledge of diagnostic compounds and their associated degradation products formed during vessel use or burial (Evershed, 2008a, Evershed, 2008b). A variety of plant products have been identified through this method, including leafy vegetables, resins and kernel oil (Evershed et al., 1991, Copley et al., 2001). However, detecting cereal residues from archaeological contexts remains particularly challenging due to their low lipid yield, low water-extractability and the lack of well-characterized biomarkers. Additionally, their contributions to residues may be masked by the presence of co-deposited animal fats present at several orders of magnitude higher concentrations (Hammann and Cramp, 2018). One exception is miliacin (olean-18-en-3β-ol), a pentacyclic triterpenoid methyl ether (PTME) enriched in broomcorn millet (*Panicum miliaceum;* hereafter referred to as 'millet') grains. First identified as a biomarker of millet cultivation in lake sediments, miliacin has subsequently been used to identify millet processing in archaeological pottery (Fig. 1; Jacob et al., 2008, Heron et al., 2016). Although other *Panicum* species also produce miliacin, it occurs in varying but much lower abundance than in millet, and is typically accompanied by high abundance of amyrin methyl esters, supporting its diagnostic value for millet when present in elevated concentrations in lipid residues (Bossard et al., 2013).

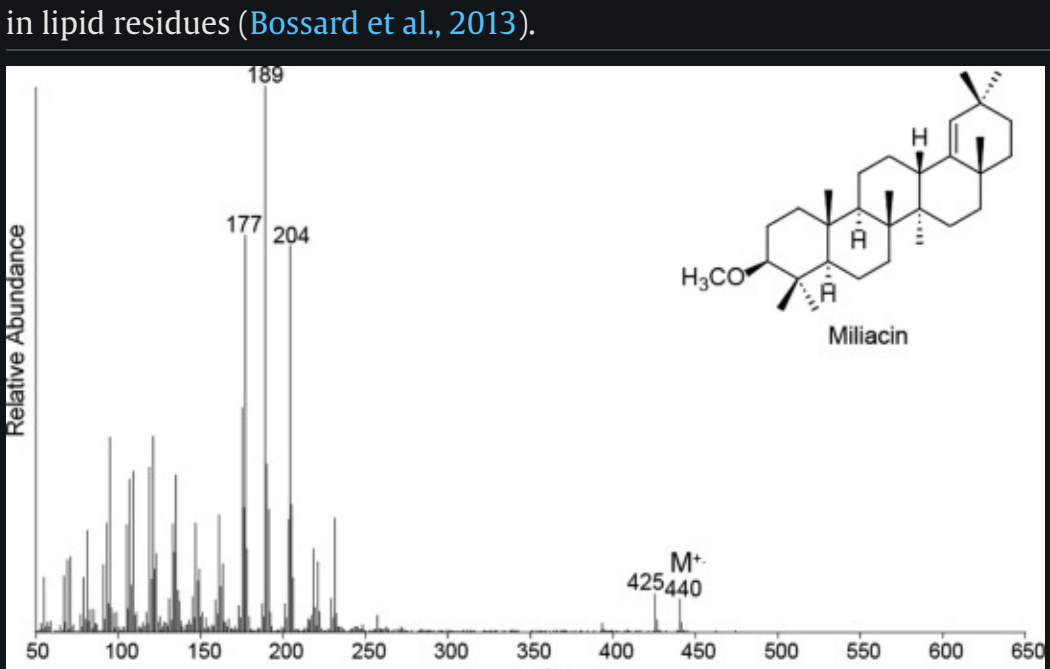

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Fig. 1. Mass spectrum of miliacin (olean-18-en-3β-ol methyl ether) extracted from modern broomcorn millet and its chemical structure.

Millet is a fast-growing, warm-season and drought-resistant cereal appearing in northern China at 8,300 BCE and domesticated by 6,500–5,500 BCE (Lu et al., 2009, Liu et al., 2016, Zhao et al., 2021, Stevens et al., 2024). Millet subsequently spread to Korea, Russia, Central Asia and Europe (Jones et al., 2011; Lightfoot et al., 2013; Stevens and Fuller, 2017, Filipovic et al., 2020, Sergusheva et al., 2022, Stevens et al., 2022). From its earliest domestication in China to its translocation across Eurasia, millet played a significant role in prehistoric subsistence strategies, providing stable starch resources and driving economic innovations in many societies. To assess the exploitation of millet and its cross-continental spread in Eurasia, organic residue analyses of pottery targeting on miliacin have been conducted on various prehistoric cultures (e.g. Heron et al., 2016, Ganzarolli et al., 2018, Shoda et al., 2018, Rageot et al., 2019; Murakami et al., 2022, Stevens et al., 2024). These studies have reported that miliacin in lipid residues is commonly present in low concentrations, and co-occurs with more abundant degraded acyl lipids dominated by fatty acids, $C_{16:0}$ and $C_{18:0}$. The presence of these lipids of contrasting sources and structures raises questions concerning the interpretation of the contributing components of such mixed residues due to: (i) the potential for differential relative release and deposition rates of lipids from cereals versus meat/dairy products, and (ii) the potentially contrasting survival rates of triterpenoid versus acyl lipids during burial.

Experiments involving boiling millet grains in water in replica unglazed vessels have shown that miliacin is mobilised and absorbed into unglazed pottery vessels, although concentrations detectable were low (Heron et al., 2016). Beyond this our understanding of the liberation of lipids from millet grains and absorption into the pottery matrix when processed sequential or concurrently with other foodstuffs is limited. Further, we have no understanding of how burial impacts miliacin survival relative to co-deposited acyl lipids, and thus, the extent to which archaeological lipid extracts reflect the compositions of original ingredients. This is particularly important in cases where millet was processed as a minor food constituent, as it may lead to an underestimation of the extent of millet processing.

To address these questions, millet and mixtures of millet with meat and milk were cooked multiple times in replica unglazed pottery vessels to evaluate the absorption of the different lipids into the ceramic matrix. Additionally, potsherds dosed with millet lipids or taken from cooking vessels used to cook millet and other foodstuffs, were submitted to laboratory degradation to simulate microbial decay in the burial environment. Quantitative analyses of the lipid extracts of these sherds would allow evaluation of factors affecting millet lipid, especially miliacin, survival relative to animal fat lipids.

2. Materials and methods

2.1. Chemicals, standards and modern samples

Millet grains were obtained from Rani Foods, India. Organic milk (3.7% fat), lamb (ca. 25% fat), pork (ca. 10% fat) and beef (ca. 25% fat) meat were obtained from a local butcher in Bristol, UK. Mushroom compost was purchased from Woodland Horticulture Limited, providing a controlled, dynamic environment that closely simulates the microbial activity and chemical conditions found in natural soil systems. Cooking experiments were performed using unglazed replica pots manufactured from a mixture of clay and sand (3:1) and fired at 1000 °C (The Pot Shop, Lincoln, UK). Sand (25%) is added to the clay to increase the porosity of the replica vessels. While this firing temperature may be higher than that of prehistoric pottery, it was not the intention to produce replica pots that were identical to the archaeological vessels in all aspects. Rather, the aim was to provide a consistent ceramic fabric for controlled experiments and study the relationship between different modes of use. The pots measured ca. 10 cm in height, 11 cm in diameter, 1 cm wall thickness and had a volume of about 500 mL.

All solvents used were of HPLC grade (Rathburn Chemicals, Walkerburn, UK) and analytical grade reagents (typically ≥ 98% purity) were used throughout. Prior to use, all reusable glassware was cleaned using Decon 90 detergent (Decon Laboratories Ltd), rinsed with acetone followed by dichloromethane (DCM), oven dried and furnaced at 450 °C for 2 h. Analytical blanks were prepared alongside lipid extractions to identify potential sources of contamination; tweezers and foil were used to manipulate sherds from the replica vessels.

2.2. Cooking experiments

Replica pots were pre-heated at ca. 350 °C for 30 min to remove any organic contaminants. Modern millet grains and meat were cleaned using double-distilled water. About 80 g of millet grains (unhusked) were boiled in 250 mL of ultrapure water (produced in the laboratory) in two replica pots (Samples M−C1 and M−C2) using a propane stove. Once the water/millet mixture reached boiling, the pot was maintained at a gentle simmer (open vessel at atmospheric pressure) for ca. 1 h, corresponding to the water boiling point. The contents were stirred every 10–15 min and water lost by evaporation topped back up to the waterline. After cooking, the millet grains were removed from the pot, and the cooking was repeated 10 times with fresh grains. For the milk and millet cooking experiment (MM-C), 80 g of millet were cooked 10 times with 250 mL of milk (3.7% fat). Milk lost by evaporation was topped up to the waterline. In total 500 mL milk was used for each cooking event, hence, 5 L in total for the ten cooking events.

In three further experiments, 80 g of millet grains (wet weight) were cooked with 100 g of beef, lamb or pork meat (wet weight) in 250 mL of ultrapure water (produced in the laboratory) in three replica pots (denoted BM-C, LM-C, and PM-C, respectively) for ten repeated processing. Portions of the pots from the rim, middle, and bottom areas were removed using a hammer and a cleaned chisel, cleaned and extracted as described below.

2.3. Dosing of sherds with millet lipids

Forty grams of millet grains were freeze-dried at −51 °C for 24 h and crushed to a fine powder using a glass mortar and pestle. Millet was extracted in 20 subsamples using 2 × 10 mL chloroform/methanol 2:1 (*v*/*v*) under sonication (20 min). The solvent phase was evaporated using a gentle stream of nitrogen and re-diluted in 3.5 mL chloroform/methanol 2:1 (*v*/*v*). All total lipid extractions (TLEs) were combined and made up to 200 mL with chloroform/methanol 2:1 (*v*/*v*). An aliquot of the solution was analysed by high temperature-gas chromatography (HT-GC) and HT-GC-mass spectrometry (HT-GC–MS). The concentration of the TLE was determined to be 4.86 mg mL^{-1} (verified by HT-GC with reference to the internal standard; C_{34} *n*-alkane). Potsherds (ca. 4 g of each one; $n = 33$) from an unused replica pot were heated at 350 °C for 30 min prior to dosing to remove any potential contamination and dosed with the millet TLE by 8 × 10 min sonication. After dosing, the sherds were left to dry overnight, and three sherds were extracted to provide a Day 0 lipid concentration, while the remaining sherds were used for degradation experiments.

2.4. Design of degradation experiments

Degradation experiments were performed in a laboratory incubator at 35 °C under oxic and anoxic conditions. Sherds were placed in glass bottles containing mushroom compost to simulate field degradation during burial. For the oxic conditions, a furnaced 1 L glass bottle was quarter filled with compost. Fifteen millet lipid-dosed potsherds were placed in this bottle and covered with more compost to ensure intimate contact with compost as a source of microbes. About 20 mL of water was added to provide moisture and the bottle was closed with a bung of furnaced glass wool. Water lost through evaporation was replaced every week. Three sherds were taken from the bottle after 7, 14, 28, 70 and 140 days. For the anoxic incubations, furnaced 250 mL glass jars ($n = 5$) were half-filled compost and, with continuous purging with N_2, three millet lipid dosed potsherds were added to each jar and covered with compost. Five millilitres of water was added and the bottles tightly closed using rubber bungs and screw lids sealed with PTFE tape. One bottle containing three sherds was taken from the incubator after 7, 14, 28, 70 and 140 days.

Analogous to the laboratory decay experiments with millet-lipid dosed sherds, three sherds were taken from the rim area of the pots BM-C, PM-C and MM-C, and set-up for longer incubation periods (210 days) than pure millet lipids noting the much higher lipid concentrations in the cooking pots used for processing mixtures of foodstuffs. The sherds were taken from the incubation bottles under oxic condition after 70, 140 and 210 days.

Three blank sherds in compost and three sherds from dosing or cooking pots were incubated in jars without compost under in oxic and anoxic conditions to provide controls and blanks.

2.5. Lipid extraction of potsherds and modern samples

Millet grains and animal meat were cleaned using DCM extracted double-distilled water and freeze dried at − 51 °C for 24 h prior to extraction. Potsherd surfaces were cleaned using a modelling drill to remove any exogenous lipids (e.g. soil or finger lipids due to handling). About 2 g of ceramic fabric or 0.1 g modern samples (dry weight) were taken and ground to a fine powder using a mortar and pestle. Lipids were extracted using the solvent extraction method as described below (Evershed et al., 1990).

2.5.1. Solvent extraction

The lipid extraction protocol followed Evershed et al. (1990) whereby powdered sherd with 40 μg of added internal standard (C_{34} *n*-alkane, 40 μL of 1 mg mL^{-1} solution) were extracted by sonication (20 min × 2) using 2 × 10 mL chloroform/methanol (2:1 *v*/*v*). An aliquot of the TLEs was trimethylsilylated with 40 μL of *N*,*O*-bis(trimethylsilyl)trifluoroacetamide (BSTFA) prior to analysis by HT-GC and HT-GC–MS.

2.5.2. HT-GC analysis of lipid extracts

HT-GC analysis of trimethylsilylated TLEs was carried out using an Agilent Technologies 7890A GC fitted with an autosampler injector and flame ionisation detector (FID). The derivatised TLEs (1 μL) were introduced to the system by an on-column injector. The oven was equipped with a 15 m × 0.32 mm (Agilent J&W Scientific) fused silica capillary column coated with a 100% dimethylpolysiloxane DB-1 HT non-polar stationary phase (0.10 μm film thickness). The GC temperature programme involved an initial 50 °C hold for 2 min after injection, followed by temperature ramping at 10 °C min^{-1} up to 350 °C, and an isothermal hold for 10 min. Helium was used as the carrier gas maintained at a constant flow (4.26 mL min^{-1}).

2.5.3. HT-GC–MS analysis of lipid extracts

HT-GC–MS analyses of trimethylsilyated TLEs were performed using a ThermoScientific Trace 1300 gas chromatograph coupled to an ISQ single quadrupole mass spectrometer. Solutions were introduced *via* a PTV injector set to split mode. The oven was equipped with a 15 m × 0.32 mm (Agilent J&W Scientific) fused silica capillary column coated with a 100% dimethylpolysiloxane DB-1 HT non-polar stationary phase (0.10 μm film thickness). The GC temperature programme for was set to hold at 50 °C for 2 min, followed by a gradient ramp to 280 °C at 10 °C min^{-1} and then 380 °C at 25 °C min^{-1} before a final isothermal at 380 °C for 5 min. Helium was used as the carrier gas set to a constant flow of 5 mL min^{-1}. The MS was operated in electron ionisation (EI) mode at 70 eV, with a GC interface temperature of 380 °C and a source temperature of 340 °C. The emission current was set to 150 μA and the MS was set to acquire in the range of *m*/*z* 50–950 Daltons at 2 scans s^{-1} in full scan mode. For detection of miliacin the MS was operated under the exact same conditions as above but with selected ion monitoring (SIM) mode, scanning the ions *m*/*z* 189 and 440.

3. Results and discussion

The investigation involved five major components (i) the study of the solvent lipid composition of fresh millet grains and animal products likely to have been processed with millet in prehistory, (ii) the composition of lipids released from millet grain by boiling in water in unglazed replica cooking pots, (iii) the composition of lipids resulting from the boiling millet grains in water with animal meat and milk in replica cooking pots, (iv) the vertical distribution of lipid concentrations in replica pots used to boil millet grains in water and with animal meat and milk, and (v) the laboratory degradation of millet lipids dosed into potsherds and the cooking pots used to cook animal products and millet. The overarching aim of the experiments was to gain a better understanding of the behaviour of the miliacin biomarker relative to fatty acyl lipid biomarkers during cooking by boiling and microbial degradation during burial.

3.1. Composition of millet lipids

Lipid extraction of the freeze-dried millet grains followed by HT-GC analysis showed the presence of the anticipated lipid classes with a mean total lipid concentration of 27 mg g^{-1}. The $C_{16:0}$, $C_{18:1}$ and $C_{18:2}$ fatty acids were the major components with minor components including the $C_{18:0}$ fatty acid, C_{52} and C_{54} triacylglycerols (TAGs), C_{34} and C_{36} diacylglycerols (DAGs), miliacin, β-sitosterol and sitosteryl esters. Non-lipid compounds, notably carbohydrates (disaccharides), were also detected (Fig. 2a). The lipid extraction from the beef meat contained TAGs with acyl carbon numbers ranging from C_{46} to C_{54}, with C_{52} predominant and C_{50} in slightly lower abundance. The overall distribution was not very different for the TAG distribution of lamb, except for the predominance of the C_{52} and the absence of the C_{44} TAG in ovine fats. TAGs in the pork lipid extract ranged between C_{46} and C_{56}, with the C_{52} TAGs dominant. Fresh milk consisted of a much wider TAG distribution containing acyl carbon numbers ranging between 28 and 56, with the components numbered from C_{36} to C_{52} present in similar abundance.

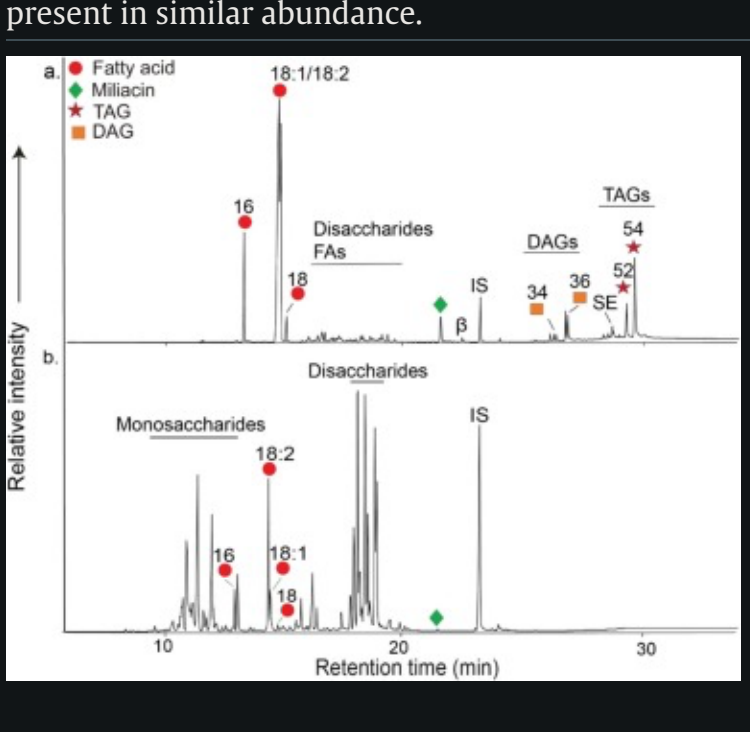

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Fig. 2. Partial (a) HT-GC profiles of the TLEs from millet grains, and (b) from the M−C1 used to boil millet alone in water for 1 h. Peak identities are: X:Y corresponds to free fatty acids with carbon number X and degree of unsaturation Y; X refers to diacylglycerols with acyl carbon number X; X refers to triacylglycerols with acyl carbon number X; represents miliacin; β refers to β-sitosterol; SE refers to β-sitosteryl fatty acyl esters; IS is the added internal standard, C_{34} *n*-alkane.

3.2. Uptake of millet lipids into a ceramic matrix

3.2.1. Cooking of millet in water

Extraction and HT-GC analysis of sherds taken from the rim, body and base areas of pots M−C1 and M−C2 indicated very little lipid incorporation into the replica pots after boiling millet for 1 h. This is likely due to the low extraction and mobilization efficiency of water for cereal lipids (Hammann and Cramp, 2018, Miller et al., 2020). The highest lipid concentration was found at the base, with a mean concentration of 8 μg g^{-1}. Non-lipid components (mono- and disaccharides) dominated the extracts, along with the presence of $C_{18:2}$ and $C_{16:0}$ fatty acids. TAGs were not observed, nor DAGs or MAGs, suggesting boiling in water was ineffective releasing these lipids from the grains. Miliacin was detected, but the concentration was very low (0.14 μg g^{-1}; Fig. 2b).

The vertical concentration profiles of M−C1 and M−C2 differed from the typical pattern observed in vessels used for cooking animal and leafy plant products (Evershed et al., 1995). A high lipid concentration was observed at the base of the vessel, with decreasing concentrations towards its top part (Fig. 3a). A similar concentration profile has been observed in pots used to cook thick or doughy contents in Africa, supporting the hypothesis that these vessels were used exclusively for boiling cereals or other thick or starch rich foods (Drieu et al., 2022). In contrast, Evershed et al. (1995) demonstrated that during the repeated boiling of lamb meat and cabbage leaf, lipids tend to accumulate in the rim areas and upper body areas of the pots.

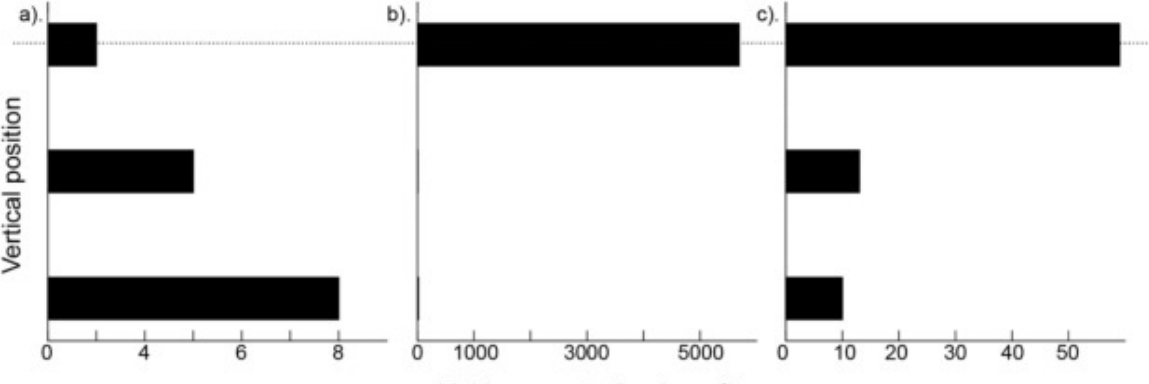

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Fig. 3. Lipid concentrations determined along the vertical profiles of pots arising from cooking millet with (a) water, (b) beef and (c) milk, respectively for 1 h after 10 times. The grey dotted line indicates the filling level for each vessel. Note the different scales of the lipid concentrations for the three treatments.

3.2.2. Composition of lipids obtained from the cooking pot used to cook animal products and millet

It has been previously observed that cooking cereals together with animal products enhanced the release of cereal lipids, resulting in a higher concentration of cereal biomarkers absorbed into the ceramic vessels (Hammann and Cramp, 2018). Given the very low concentrations of millet lipids freed from the grains in the boiling water experiments, further cooking experiments focused on the co-processing of millet with different animal products.

Lipids extracted from the replica pottery used to boil different animal products together with millet showed similar distributions of lipids to their parent animal fats with TAGs as the major components of the TLEs (Fig. 4). The highest lipid concentrations and miliacin concentrations were all found in the upper body and rim regions of the pots, with the total absorbed lipid concentration maximising at 3,787 μg g^{-1} for LM-C, 5,695 μg g^{-1} for BM-C, 2,066 μg g^{-1} for PM-C and 59 μg g^{-1} for MM-C (Fig. 3b, c; Table 1 for the results of the cooking experiments). The maximum concentration of miliacin detected from MM-C was 0.12 μg g^{-1}, which indicated no improvement in the extractability of the millet lipids from cooking milk together with millet compared to water cooking (0.14 μg g^{-1}). The uptake of miliacin into the pot used to cook millet with beef, lamb, pork was 0.82, 0.62, and 0.51 μg g^{-1}, respectively, significantly higher than in the concentrations observed when boiling millet in water alone. This enhanced miliacin recovery was consistent with previous experimental observations, that more abundant animal fats act as a carrier during cooking, increasing the efficiency of solubilisation and mobilisation of hydrophobic lipids leading to enhanced incorporation into the ceramic matrix (Hammann and Cramp, 2018). Although this enhanced incorporation may theoretically apply to both miliacin and millet acyl lipids, $C_{16:0}$ and $C_{18:0}$ FA cannot be differentiated from animal fats in such mixed residues. Therefore, the interpretation of enhanced millet input relies primarily on the detection of miliacin.

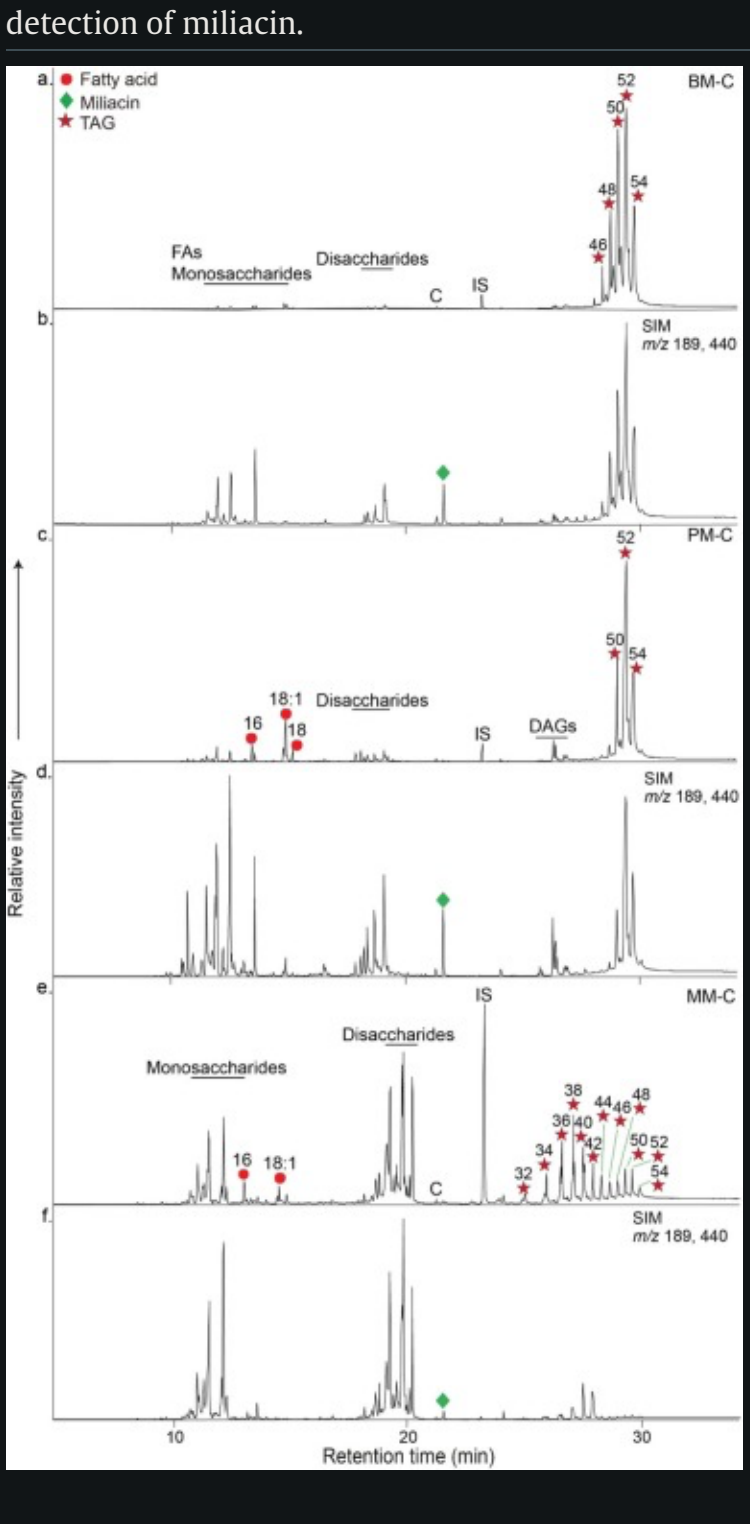

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Fig. 4. Partial HT-GC profiles of the TLEs obtained using the chloroform/methanol extraction method from the rim of the pots used to cook (a) beef, (b) pork, and (c) milk, and (b), (d), and (f) respective SIM mass chromatograms (*m*/*z* 189, 440) for miliacin. Peak identities are: X:Y corresponds to free fatty acids with carbon numbered X and degree of unsaturation Y; X refers to TAGs with carbon number X; represents miliacin; C represents cholesterol; IS is the added internal standard, C_{34} *n*-alkane.

Table 1. Summary of sample codes, descriptions, vessel parts, total lipid and miliacin concentrations (μg g^{-1}) of raw samples, the replica pots after cooking and degradation.

Sample	Description	Vessel part	Total lipid concentrations (
M−E	Modern millet extraction		27,000
M−C1, M−C2	Millet boiled in water, 10 × 1h, cooking; two replica pots	Rim	3
		Mid	5
		Bottom	8
M−D	Sherds dosed with millet lipids		616
M−DA	M−D buried under anoxic conditions for 140 days		145
M−DO	M−D buried under oxic conditions for 140 days		15
B-E	Raw beef extraction		–
BM-C	Beef + millet boiled in water, 10 × 1h	Rim	5695
		Mid	6
		Bottom	19
BM-DO	Rim sherds from BM-C, buried under oxic conditions for 210 days		479
L-E	Raw lamb extraction		–
LM-C	Lamb + millet boiled in water, 10 × 1h	Rim	3787
		Mid	12
		Bottom	274
P-E	Raw pork extraction		–
PM-C	Pork + millet boiled in water, 10 × 1h	Rim	2066
		Mid	3
		Bottom	11
PM-DO	Rim sherds from PM-C, buried under oxic condition for 210 days		147
MM-E	Milk extraction		–
MM-C	Milk + millet boiled in water, 10 × 1h	Rim	59
		Mid	13
		Bottom	10
MM-DO	Rim sherds from MM-C, buried under oxic condition for 210 days		5

Although limited concentrations of miliacin (up to 0.82 μg g^{-1}) were absorbed into the pot matrix during cooking experiments, even with animal meat products, this does not exclude its potential recovery in high concentrations from archaeological potsherds. The laboratory experiments only involved 10 cooking events, while the actual use of vessels in prehistory would likely have involved repeated cooking over many years, accumulating significantly higher miliacin concentrations. To account for potentially higher miliacin recovery and investigate lipid degradation behaviour, sherds were dosed with millet-TLE solutions and compared with those taken from the rim area of the pots used to cook beef, pork and milk in the degradation experiments discussed below.

3.3. Laboratory decay experiments

3.3.1. Decay of pure millet lipids absorbed in pot sherds

Dosing blank sherds with millet-TLE solutions resulted in a concentration of 616 μg g^{-1} lipid absorbed into the sherd matrix. The lipid composition in the sherds corresponded well with that of the TLE of the reference millet grains.

During laboratory decay of millet lipids under oxic conditions, DAGs and TAGs were hydrolysed to free fatty acids, resulting in an increase of free fatty acids during the first 7 days. A 65% decrease in the lipid concentration was observed by day 14, with a 90% decrease of total lipids occurring by day 28 (73 μg g^{-1}; Fig. 5), with the concentration remaining low thereafter. The major influence on this pattern of degradation was a substantial decrease of the concentration of unsaturated fatty acids, $C_{18:1}$ and $C_{18:2}$, probably because of their susceptibility to microbial degradation or loss through oxidation (Dudd et al., 1998, Ailaud, 2002). After 140 days under oxic conditions, TAGs and fatty acids were almost completely degraded, resulting in a lipid profile consisting almost exclusively of miliacin and β-sitosterol (Fig. 6a).

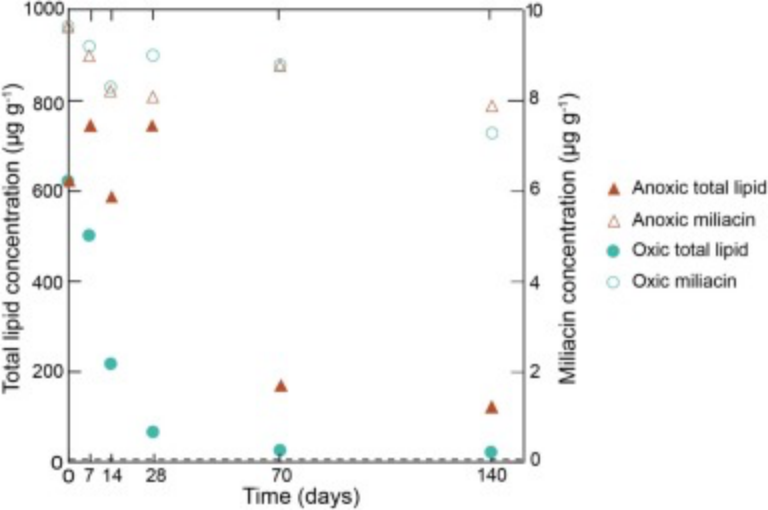

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Fig. 5. Changes in the total lipid and miliacin concentrations (µg g^{-1}) under oxic (,) and anoxic (,) conditions during the experimental decay of millet lipids absorbed in sherds. The dotted line represents reliable interpretable total lipid concentrations of 5 µg g^{-1} (Evershed, 2008a).

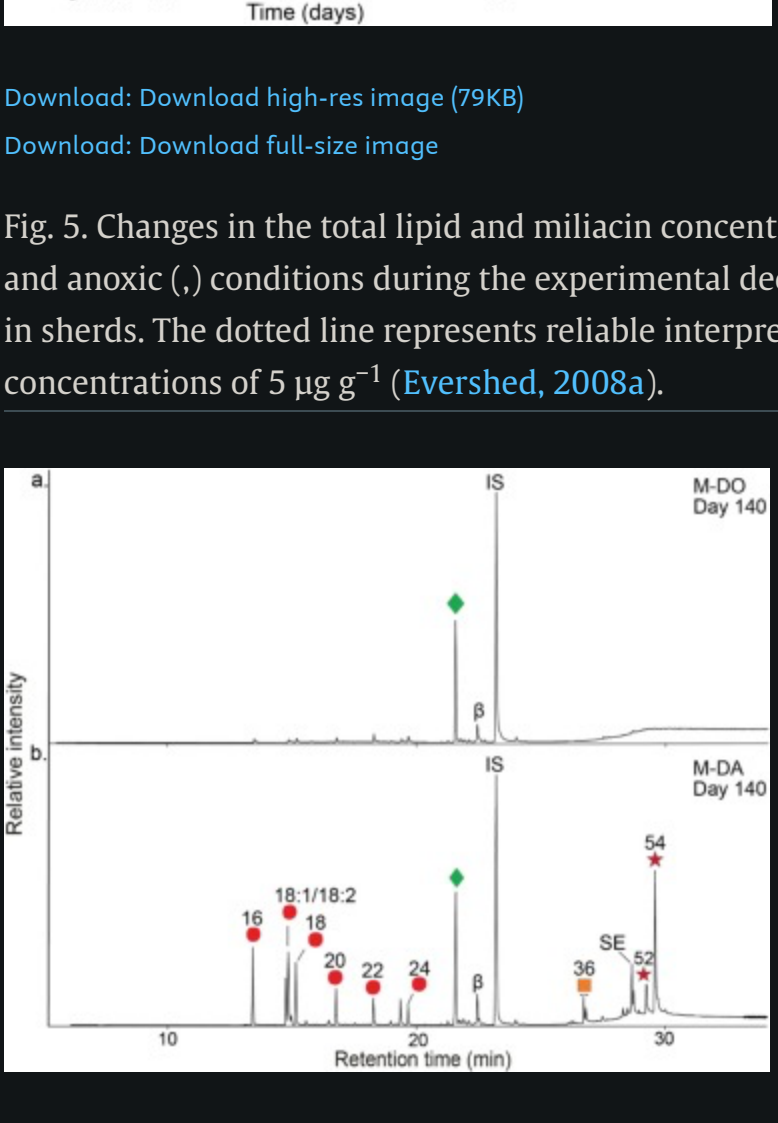

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Fig. 6. Partial HT-GC profiles of the TLEs of millet lipid degradation (a) under oxic and (b) anoxic conditions after 140-day incubations. Peak identities are: X:Y corresponds to free fatty acids with carbon number X and degree of unsaturation Y; X refers to DAGs with carbon number X; X refers to TAGs with carbon number X; represents miliacin; β refers to β-sitosterol; SE refers to sitosteryl fatty acyl esters; IS is the added internal standard, C_{34} *n*-alkane.

As anticipated, the anoxic degradation occurred much slower than oxic degradation. After 4 weeks the lipid content was relatively unaltered. The lipid concentration decreased to 176 µg g^{-1} by day 70, and 145 µg g^{-1} remaining after 140 days (Fig. 5). The composition of the lipids after degradation included free fatty acids, $C_{16:0}$, $C_{18:1}$, $C_{18:2}$ and $C_{18:0}$, DAGs and TAGs, closely resembling that dosed into the sherds, despite changes in the relative proportion of each component (Fig. 6b). Long-chain fatty acids increased in relative abundance compared to the other components after 140 days of anoxic incubation compared to the start of the experiment. These results indicate that the anoxic conditions are more favourable for lipid survival, aligning with results from previous decay experiments involving other commodities (Dudd et al., 1998, Evershed, 2008a, Hammann and Cramp, 2018).

β-Sitosterol proved to be resilient to microbial degradation in the experiments. The starting concentration in the dosed sherds was 1.8 µg g^{-1}, with levels in the sherds still at 0.8 µg g^{-1} and 1.6 µg g^{-1} after 140 days of oxic and anoxic degradation, respectively. β-Sitosterol was virtually absent from the TLEs of cooking pots and archaeological cooking pots, which agrees with the finding that sterols are readily degraded during cooking (Hammann et al., 2018).

3.3.2. Stability of miliacin compared to fatty acyl lipids

The initial concentration of miliacin was 9.6 µg g^{-1} in the sherds dosed with millet lipids. After 140 days of decay under oxic and anoxic conditions the concentration of miliacin remaining was 7.7 and 7.9 µg g^{-1}, respectively, showing that it is resistant to microbial degradation (Fig. 5). Noteworthy, whereas ca. 80% of miliacin survived after 140 day under oxic degradation, the acyl lipids dropped to 5% of the initial concentration, with miliacin becoming the most abundant component (Fig. 6a). Under anoxic conditions, the ratio of the concentration of FA $C_{16:0}$ compared to miliacin (FA $C_{16:0}$/miliacin; 0.95) and FA $C_{18:0}$/miliacin (0.7) after the decay were much lower than those observed for fresh millet grain lipids (6.4 and 1.6, respectively). This indicated the concentration of miliacin was steady, indicating it is more resistant to degradation than fatty acyl lipids, accounting for the elevated miliacin concentrations seen in many archaeological residues compared to fatty acyl lipids. The similar concentrations of miliacin after degradation under oxic and anoxic conditions indicates that these two burial conditions have limited influence on its survival in archaeological pottery.

3.3.3. Decay trajectories of mixtures of animal and millet lipids

The gas chromatograms of the TLEs obtained from the replica pots used to cook different animal products and millet grains (Fig. 4) were dominated by TAGs, the main lipid components of parent animal fats. After 210 days of burial under oxic conditions the total lipid concentration was 479, 147, and 5 µg g^{-1}, respectively, for BM-C, PM-C, and MM-C. The major components seen were free fatty acids, $C_{16:0}$, $C_{18:0}$ and $C_{18:1}$. Only traces of TAGs, DAGs and MAGs, were observed, suggesting they had been hydrolysed to free fatty acids (Fig. 7). Miliacin was seen only in trace concentration but was readily detectable by GC–MS under selected ion monitoring (SIM) mode (Fig. 7b, d). There was no appreciable extractable lipid remaining in the pot used to cook milk and millet after 210 days of incubation under oxic conditions. However, miliacin was detected using SIM (Fig. 7f), which emphasises again that miliacin is more resistant to microbial degradation than fatty acyl lipids.

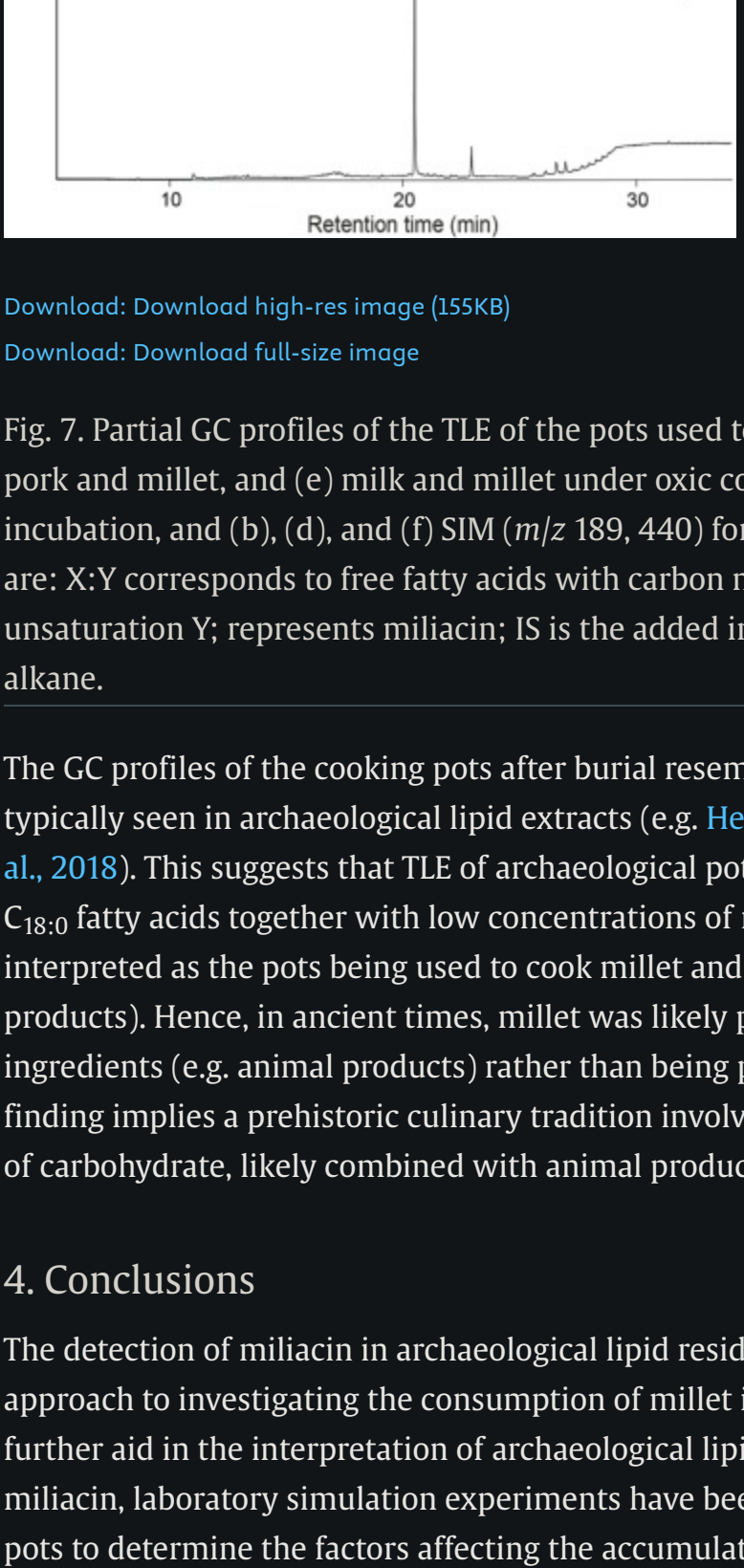

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Fig. 7. Partial GC profiles of the TLE of the pots used to cook (a) beef and millet, (c) pork and millet, and (e) milk and millet under oxic conditions after 210 day incubation, and (b), (d), and (f) SIM (m/z 189, 440) for miliacin. Peak identities are: X:Y corresponds to free fatty acids with carbon number X and degree of unsaturation Y; represents miliacin; IS is the added internal standard, C_{34} *n*-alkane.

The GC profiles of the cooking pots after burial resembled the distribution typically seen in archaeological lipid extracts (e.g. Heron et al., 2016, Ganzarolli et al., 2018). This suggests that TLE of archaeological potsherds containing $C_{16:0}$ and $C_{18:0}$ fatty acids together with low concentrations of miliacin can be reliably interpreted as the pots being used to cook millet and animal meat (or dairy products). Hence, in ancient times, millet was likely processed with other ingredients (e.g. animal products) rather than being prepared on its own. This finding implies a prehistoric culinary tradition involving millet as a staple source of carbohydrate, likely combined with animal products, in prehistoric diets.

4. Conclusions

The detection of miliacin in archaeological lipid residues provides a promising approach to investigating the consumption of millet in prehistory. Herein, to further aid in the interpretation of archaeological lipid residues containing miliacin, laboratory simulation experiments have been carried out using replica pots to determine the factors affecting the accumulation of miliacin during vessel use and its subsequent decay during burial. These experimental investigations presented have provided a number of new insights into the lipid residues associated with the processing of broomcorn millet, specifically:

1. In contrast to cooking animal meat or leaf vegetables within pottery vessels, the highest lipid concentrations of the pots used to boil millet were found in the bottom parts, which appears to be a specific pattern of lipid accumulation resulting from boiling cereals exclusively in water.
2. Cooking mixtures of animal products and millet produced a lipid distribution dominated by the parent animal products.
3. Due to the high fat content of meat co-processing of meat and millet improves the liberation and absorption of miliacin into the pot walls during vessel use.
4. Laboratory decay of the sherds dosed with millet lipids showed miliacin has a higher survival rate than fatty acyl lipids and oxic/anoxic burial conditions have no strong influence on miliacin preservation.
5. The decay experiments revealed that if the vessel is used solely for millet processing, its lipid residue after burial will be dominated by miliacin.
6. The detection of $C_{16:0}$ and $C_{18:0}$ fatty acids together with miliacin in lipid extracts of archaeological cooking vessels indicates the processing of animal products and millet in the same vessels.

These insights provide a robust experimental foundation for interpreting millet-based culinary practices in archaeological pottery, especially alongside animal products, and strengthen the use of miliacin as a biomarker for millet processing in prehistoric contexts.

CRediT authorship contribution statement

Mengyao Zhang: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation. **Mélanie Roffet-Salque:** Writing – review & editing, Resources. **Lucy J.E. Cramp:** Writing – review & editing, Methodology. **Richard P. Evershed:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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